

The density Hajnal–Szemerédi theorem

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Abstract

The Corrádi–Hajnal and Hajnal–Szemerédi theorems determine exact minimum-degree thresholds for packing vertex-disjoint cliques. The corresponding edge-density problem is known for triangles and, asymptotically, for cliques of size four. For every fixed integer $r \geq 3$, we determine asymptotically the maximum number of edges in an n -vertex graph that contains no $k + 1$ vertex-disjoint copies of K_r , uniformly over the full range $0 \leq k \leq n/r$. Our result recovers the known density formulas for K_3 and K_4 and proves the $o(n^2)$ -error form of the conjecture of Hou, Hu, Liu, and Zhang for every $r \geq 5$.

The proof proceeds through a weighted clique-tiling theorem whose induction starts with the Erdős–Gallai matching theorem. A support-minimal weighted counterexample has the property that the quadratic form associated with its adjacency matrix is strictly negative on a subspace of codimension at most two. Spectral interlacing and a sign rule on induced paths force its support to be either complete or a clique blow-up of P_3 . Extreme fractional covers on the latter support have at most two positive levels, and the remaining case reduces to a positive-semidefinite quadratic form. Graphon tiling-cover duality then transfers the weighted result to finite graphs.

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1 Introduction

1.1 Motivation and main result

For graphs F and G , an F -tiling in G is a collection of pairwise vertex-disjoint copies of F . We denote the maximum size of such a tiling by $\nu(F, G)$. For integers $n, k \geq 0$, let

$$\text{ex}(n, (k+1)F)$$

be the maximum number of edges in an n -vertex graph G satisfying $\nu(F, G) \leq k$. Thus the classical Turán problem is the case $k = 0$, whereas $F = K_2$ is the matching problem solved by Erdős and Gallai [EG59].

The minimum-degree version of the clique-tiling problem is governed by the theorems of Corrádi–Hajnal and Hajnal–Szemerédi [CH63, HS70]. The edge-density version is more delicate: even for a fixed clique size, different constructions are extremal in different ranges of k . Early results treated small values of k and the range near a perfect clique tiling [Erd62, Moo68, AF85, BKT13]; related partial-tiling problems were studied, for example, in [GH12]. Allen, Böttcher, Hladký, and Piguet determined the triangle case [ABHP15]. Hou, Hu, Liu, and Zhang later determined the asymptotic answer for K_4 and proposed $r+1$ effective extremal families for every $r \geq 5$ [HHLZ25].

We prove the density statement for every clique size in a single framework. For $r = 3$ and $r = 4$, our $o(n^2)$ error is weaker than the known $O(n)$ error, but the same weighted argument also supplies the induction for every higher clique size. For $r \geq 5$, the result proves the density form of the conjecture in [HHLZ25]. The stronger linear error remains outside the scope of the present paper.

We now define the candidate density. Fix an integer $r \geq 3$ and $x \in [0, 1/r]$. For $1 \leq i \leq r-1$, let

$$B_{r,i}(x) := \frac{1}{2} \left(1 - \frac{(1-ix)^2}{r-i} \right). \quad (1.1)$$

Define

$$A_r(x) := \frac{r-2}{2(r-1)} + 2x^2, \quad 0 \leq x \leq \frac{1}{2(r-1)}, \quad (1.2)$$

and, writing $\ell = 1 - rx$,

$$R_r(x) := \frac{1}{2} - \ell + \frac{r-1}{2} \ell^2, \quad \frac{r-2}{r(r-1)} \leq x \leq \frac{1}{r}. \quad (1.3)$$

Curves outside their displayed domains are omitted from maxima. Put

$$F_r(x) := \max \left(\{B_{r,i}(x) : 1 \leq i \leq r-1\} \cup \{A_r(x)\} \cup \{R_r(x)\} \right). \quad (1.4)$$

For the weighted induction base, set

$$F_2(x) = \max \left\{ x - \frac{x^2}{2}, 2x^2 \right\}, \quad 0 \leq x \leq \frac{1}{2}. \quad (1.5)$$

Theorem 1.1. *For every fixed integer $r \geq 3$, uniformly for integers $0 \leq k \leq n/r$,*

$$\text{ex}(n, (k+1)K_r) = F_r(k/n)n^2 + o_r(n^2).$$

Remark 1.2. (i) *The uniformity in [Theorem 1.1](#) means that the error term is bounded by $\eta_r(n)n^2$ for a function $\eta_r(n) \rightarrow 0$, independently of the choice of $k \in \{0, \dots, \lfloor n/r \rfloor\}$. For $r = 3$ and $r = 4$, the explicit piecewise forms of F_r are recorded in [Section 8](#).*

(ii) *The previously known results for K_3 and K_4 have an $O(n)$ error. Our argument is designed to treat every r uniformly and does not recover this sharper finite error term. Its new content is the upper bound for all $r \geq 5$.*

1.2 Extremal constructions

We next describe the constructions that give the lower bound. For this purpose, a *fractional K_r -cover* of a graph G is a nonnegative weight function $c : V(G) \rightarrow \mathbb{R}$ such that $\sum_{v \in Q} c_v \geq 1$ for every copy $Q \cong K_r$ in G . If such a cover has total weight at most k , then every K_r -tiling in G has size at most k . The three families below all admit a cover of total weight k .

The B -constructions. Fix $1 \leq i \leq r-1$. Take a clique X of size ik , partition the remaining $n - ik$ vertices as equally as possible into $r - i$ independent classes, join distinct outside classes completely, and join X completely to the outside. Every K_r uses at least i vertices of X , so assigning weight $1/i$ to X and zero elsewhere is a cover of total weight k . Writing $x = k/n$, the number of edges is

$$\binom{ik}{2} + ik(n - ik) + \frac{1}{2} \left(1 - \frac{1}{r-i}\right) (n - ik)^2 + O_r(n) = B_{r,i}(x)n^2 + O_r(n).$$

The A -construction. Assume $k/n \leq 1/[2(r-1)]$. Start with a balanced complete $(r-1)$ -partite graph on n vertices, choose a set X of size $2k$ inside one part, and add all edges inside X . Every K_r uses at least two vertices of X , so weight $1/2$ on X is a cover of total weight k . Writing $x = k/n$, the edge count is

$$\left(1 - \frac{1}{r-1}\right) \frac{n^2}{2} + \binom{2k}{2} + O_r(n) = A_r(x)n^2 + O_r(n).$$

The more general curves

$$A_{r,i}(x) = \frac{r-i}{2(r-i+1)} + \frac{i^2 x^2}{2}$$

from [\[HHLZ25\]](#) are dominated by $A_r = A_{r,2}$ whenever $i \geq 3$.

Lemma 1.3. *For $r \geq 4$, $3 \leq i \leq r-1$, and $0 \leq x \leq 1/[i(r-i+1)]$,*

$$A_{r,i}(x) \leq A_r(x).$$

Proof. A direct subtraction gives

$$A_r(x) - A_{r,i}(x) = \frac{i-2}{2} \left(\frac{1}{(r-1)(r-i+1)} - (i+2)x^2 \right).$$

For these parameters, $i(r-i+1) \geq 2(r-1)$, so A_r is defined throughout the domain under consideration. The domain assumption and

$$i^2(r-i+1) - (i+2)(r-1) = (i-2)((i+1)(r-i) - 1)$$

show that the expression in parentheses is nonnegative. □

The R -construction. Assume $k/n \geq (r-2)/[r(r-1)]$ and put $L = n - rk$. Take outside classes $V_1, \dots, V_{r-1}$ of size L and a clique

$$X \text{ of size } a = n - (r-1)L.$$

Put $Z_1 = V_1$, and put $Y_j = V_j$ for $2 \leq j \leq r-1$. Place a complete $(r-1)$ -partite graph on the outside classes, and join X completely to every Y_j but not to Z_1 . No K_r meets Z_1 , and

$$|X| + \sum_{j=2}^{r-1} |Y_j| = a + (r-2)L = rk.$$

Thus weight $1/r$ on $X \cup \bigcup_{j=2}^{r-1} Y_j$ is a cover of total weight k . Writing $x = k/n$, the edge count is

$$\binom{a}{2} + a(r-2)L + \binom{r-1}{2}L^2 = R_r(x)n^2 + O_r(n).$$

Proposition 1.4. *For every fixed $r \geq 3$, uniformly for $0 \leq k \leq n/r$,*

$$\text{ex}(n, (k+1)K_r) \geq F_r(k/n)n^2 - O_r(n).$$

Proof. Use the B , A , or R construction corresponding to a curve attaining the maximum in (1.4). The covers described above prove that the K_r -matching number is at most k , and the displayed edge calculations give the estimate. $\square$

Lemma 1.5. *For every $r \geq 3$, the function F_r is nondecreasing and continuous on $[0, 1/r]$. The same is true of F_2 on $[0, 1/2]$.*

Proof. The assertion for F_2 is immediate. Every $B_{r,i}$ is increasing on $[0, 1/r]$, and A_r is increasing on its domain. Moreover,

$$R'_r(x) = r(1 - (r-1)(1 - rx)) \geq 0$$

on the domain of R_r . At the right endpoint of the A_r domain,

$$A_r\left(\frac{1}{2(r-1)}\right) = B_{r,2}\left(\frac{1}{2(r-1)}\right).$$

At the left endpoint $x_R = (r-2)/[r(r-1)]$ of the R_r domain,

$$B_{r,r-1}(x_R) - R_r(x_R) = \frac{(r-2)^2}{2r^2(r-1)} > 0.$$

These endpoint comparisons show that the restricted-domain maximum is continuous at both transition points. $\square$

1.3 Outline of the proof

The lower bound in Theorem 1.1 is Proposition 1.4; the rest of the paper is devoted to the upper bound. We now describe the six main steps of the argument.

- (i) *Weighted formulation and clique-size recursion.* We formulate a finite weighted density theorem and start its induction at $r = 2$ with the Erdős–Gallai matching theorem. The Bellman peeling operator transfers the result from clique size $r-1$ to clique size r whenever the support has a universal atom of cover weight zero. The only exceptional calculations occur at $r = 3$ and $r = 4$.
- (ii) *A support-minimal counterexample.* Assuming that the weighted theorem fails, we choose a counterexample with minimal positive support and maximum edge density, and then an extreme minimum fractional cover c . For every nonzero vector z satisfying $z \perp \mathbf{1}, c$, the adjacency form obeys

$$z^\top A(H)z < 0.$$

Thus $A(H)$ has at most two nonnegative eigenvalues.

- (iii) *The co-chain reduction.* Spectral interlacing implies that $\overline{H}$ is a chain graph. On every induced path $a - b - c - d$, strict negativity gives

$$(c_c - c_a)(c_d - c_b) > 0.$$

These signs propagate along the Ferrers staircase. Edge saturation at its far corners then rules out every staircase with at least two steps. Consequently, H is either complete or a clique blow-up of P_3 .

- (iv) *Complete supports and zero atoms.* Complete supports are solved by a hypersimplex load decomposition. On a P_3 -support, a zero-cover atom in the universal module invokes the clique-size recursion, whereas a terminal zero set is handled by a weighted extension of Turán's theorem.
- (v) *Extreme-cover rigidity.* In the remaining P_3 case, an extreme minimum cover has at most two positive levels. Each of the two complete sides contributes at most one independent tight-set equation. Together with saturation across the missing end-to-end pairs, this leaves only two nonterminal cover types.
- (vi) *The final optimization and transfer.* The symmetric type gives a B -curve directly. In the second type, the Karush–Kuhn–Tucker equations reduce the masses to two variables, and the desired inequality is an explicit positive-semidefinite quadratic form. The weighted induction is then complete. Finally, graphon tiling–cover duality and lower semicontinuity transfer the result to finite graphs.

Organization of the paper. In [Section 2](#) we introduce the weighted problem, prove the $r = 2$ base case, and establish the clique-size recursion. In [Section 3](#) we analyze the spectrum and co-chain representation of a support-minimal counterexample. Sections [4](#) and [5](#) then treat complete supports, zero atoms, and extreme covers on the remaining P_3 -support. The induction is completed in [Section 6](#). In [Section 7](#) we pass from weighted graphs to graphons and finite graphs. The known formulas for K_3 and K_4 are recovered in [Section 8](#); the appendix records the two algebraic calculations used in the recursion and in the final rigid case.

2 Preliminaries and the weighted formulation

This section introduces the weighted language used throughout the proof. We then state the weighted density theorem, establish its $r = 2$ base case, and verify the Bellman recursion for the candidate functions.

2.1 Weighted tiling–cover duality

A *weighted graph* is a pair (H, p) , where H is a finite simple graph and $p = (p_v)_{v \in V(H)}$ is a probability vector with $p_v > 0$ for every v . We call H the *support graph*, refer to its vertices as *atoms*, and call p_v the mass of v . For $S \subseteq V(H)$, write $p(S) = \sum_{v \in S} p_v$; the analogous notation is used for other vectors on $V(H)$. Define

$$e(H, p) = \sum_{uv \in E(H)} p_u p_v = \frac{1}{2} p^\top A(H) p.$$

Let $\mathcal{K}_r(H)$ be the set of copies of K_r in H .

A weighted fractional K_r -tiling is a vector $t = (t_K)_{K \in \mathcal{K}_r(H)}$ satisfying

$$t_K \geq 0, \quad \sum_{K \ni v} t_K \leq p_v \quad (v \in V(H)).$$

Its size is $\sum_K t_K$, and its maximum is denoted $\text{til}_r(H, p)$. A fractional K_r -cover is a vector $c \in \mathbb{R}_{\geq 0}^{V(H)}$ satisfying

$$\sum_{v \in K} c_v \geq 1 \quad (K \in \mathcal{K}_r(H)),$$

and its mass is $\langle p, c \rangle = \sum_v p_v c_v$. Let $\text{fcov}_r(H, p)$ be the minimum cover mass. Finite-dimensional linear-programming duality gives

$$\text{til}_r(H, p) = \text{fcov}_r(H, p). \tag{2.1}$$

The constant vector $1/r$ is a cover, so both quantities lie in $[0, 1/r]$.

Theorem 2.1 (Weighted density theorem). *Let $r \geq 2$, let (H, p) be a weighted graph, and put*

$$x = \text{fcov}_r(H, p).$$

Then

$$e(H, p) \leq F_r(x),$$

where F_2 is defined in (1.5). Consequently, for every fractional K_r -cover c with $\langle p, c \rangle \leq 1/r$,

$$e(H, p) \leq F_r(\langle p, c \rangle).$$

Proposition 2.2 (Weighted Erdős–Gallai base). *Theorem 2.1 holds for $r = 2$.*

Proof. Put $x = \text{til}_2(H, p) = \text{fcov}_2(H, p)$. First suppose that all masses are rational. Fix a common denominator N_0 , let $N = tN_0$, and blow up each vertex v to an independent class of size Np_v , joining two classes completely exactly when their support vertices are adjacent. Let the resulting graph be G_N . Then

$$\frac{e(G_N)}{N^2} = e(H, p).$$

Every integral matching in G_N , divided by N and grouped according to support edges, is a weighted fractional matching in (H, p) . Hence, with $m_N = \nu(G_N)$,

$$\frac{m_N}{N} \leq x.$$

The Erdős–Gallai matching theorem [EG59] gives

$$e(G_N) \leq \max \left\{ \binom{2m_N + 1}{2}, m_N(N - m_N) + \binom{m_N}{2} \right\}.$$

After division by N^2 and passage to a subsequence, write $m_N/N \rightarrow y \leq x$. Letting $t \rightarrow \infty$ gives

$$e(H, p) \leq \max \left\{ 2y^2, y - \frac{y^2}{2} \right\} \leq F_2(x),$$

since both functions are increasing on $[0, 1/2]$. General masses follow by approximation with positive rational probability vectors and continuity of the finite linear program and the edge density. $\square$

2.2 The peeling operator

The following one-variable operator records the density obtained after peeling off a universal zero-cover atom and renormalizing the remaining support. Its dynamic-programming form is the reason for the term *Bellman recursion*.

For a function $f : [0, 1/(r-1)] \rightarrow \mathbb{R}$, define

$$(\mathcal{T}_r f)(x) := \sup_{\substack{(r-1)x \leq q \leq 1 \\ q > 0}} \{q(1-q) + q^2 f(x/q)\}. \quad (2.2)$$

When $x = 0$, one may equivalently adjoin the term $q = 0$ with value zero.

Lemma 2.3 (Clique-size recursion). *For every $r \geq 3$ and $x \in [0, 1/r]$,*

$$\mathcal{T}_r F_{r-1}(x) \leq F_r(x).$$

Proof. It is enough to treat each branch of F_{r-1} separately.

For $1 \leq i \leq r-2$,

$$\begin{aligned} & q(1-q) + q^2 B_{r-1,i}(x/q) \\ &= q - \frac{q^2}{2} - \frac{(q-ix)^2}{2(r-1-i)}. \end{aligned} \quad (2.3)$$

This is concave in q , with maximum at

$$q_i^* = \frac{r-i-1+ix}{r-i}.$$

Since

$$q_i^* - (r-1)x = \frac{(r-i-1)(1-rx)}{r-i} \geq 0, \quad 1 - q_i^* = \frac{1-ix}{r-i} \geq 0,$$

the point is feasible, and substitution gives

$$\sup_q (q(1-q) + q^2 B_{r-1,i}(x/q)) = B_{r,i}(x). \quad (2.4)$$

At the lower endpoint $q = (r-1)x$, the value with a complete remaining support is

$$q(1-q) + \frac{q^2}{2} = B_{r,r-1}(x). \quad (2.5)$$

For $r = 3$, the other branch of F_2 is $2y^2$, and

$$q(1-q) + 2x^2 = q - q^2 + 2x^2.$$

If $x \leq 1/4$, its maximum over $q \geq 2x$ occurs at $q = 1/2$ and equals $A_3(x)$. If $1/4 \leq x \leq 1/3$, the maximum occurs at $q = 2x$ and equals $B_{3,2}(x)$. Thus the recursion holds for $r = 3$.

Now let $r \geq 4$. On the A_{r-1} branch, the domain requires $q \geq 2(r-2)x$, and

$$q(1-q) + q^2 A_{r-1}(x/q) = q - \frac{r-1}{2(r-2)} q^2 + 2x^2. \quad (2.6)$$

The unconstrained maximum occurs at $q = (r-2)/(r-1)$. If $x \leq 1/[2(r-1)]$, this point is feasible and the value is $A_r(x)$. If

$$\frac{1}{2(r-1)} < x \leq \frac{1}{2(r-2)},$$

the maximum is attained at $q = 2(r-2)x$, where

$$A_{r-1}\left(\frac{1}{2(r-2)}\right) = B_{r-1,2}\left(\frac{1}{2(r-2)}\right),$$

so it is controlled by (2.4). If $x > 1/[2(r-2)]$, the branch is absent.

For $r = 4$, it remains to transform R_3 . Its domain is

$$3x \leq q \leq \min\{6x, 1\},$$

and

$$\Psi_x(q) = q(1-q) + q^2 R_3(x/q) = (1-3x)q - \frac{1}{2}q^2 + 9x^2.$$

The stationary point is $q_* = 1-3x$. If $0 < x \leq 1/9$, then $q_* \geq 6x$, so the maximum occurs at $q = 6x$. Since

$$R_3(1/6) = \frac{1}{4} < \frac{47}{144} = B_{3,1}(1/6),$$

this value is at most the transformed $B_{3,1}$ maximum, namely $B_{4,1}(x)$. If $1/9 \leq x \leq 1/6$, then q_* is feasible and

$$\Psi_x(q_*) = \frac{1}{2} - 3x + \frac{27}{2}x^2.$$

Moreover,

$$A_4(x) - \Psi_x(q_*) = -\frac{1}{6} + 3x - \frac{23}{2}x^2.$$

This difference is concave and has endpoint values $2/81$ and $1/72$, so it is nonnegative. If $1/6 \leq x \leq 1/4$, the maximum is attained at $q = 3x$ and equals $B_{4,3}(x)$. Hence the recursion holds for $r = 4$.

Finally let $r \geq 5$. On the R_{r-1} branch, direct expansion gives a quadratic polynomial in q with coefficient $(r-5)/2$. It is affine for $r=5$ and convex for $r \geq 6$, so its maximum is at a domain endpoint. The lower endpoint $q = (r-1)x$ gives (2.5). If the upper endpoint is determined by the left endpoint

$$y_0 = \frac{r-3}{(r-1)(r-2)}$$

of the R_{r-1} domain, then

$$B_{r-1,r-2}(y_0) - R_{r-1}(y_0) = \frac{(r-3)^2}{2(r-2)(r-1)^2} > 0,$$

so the transformed value is controlled by (2.4). If the upper endpoint is $q=1$, then $x \geq y_0$ and

$$B_{r,r-1}(x) - R_{r-1}(x) = \frac{(1-(r-1)x)((r-1)^2x-r+3)}{2} \geq 0.$$

This completes the recursion for every $r \geq 3$. □

Lemma 2.4 (Universal zero peeling). *Let $r \geq 3$ and assume Theorem 2.1 holds for clique size $r-1$. Let (H, p) have a fractional K_r -cover c of mass x , and suppose that a vertex b is adjacent to every other vertex and satisfies $c_b = 0$. Then*

$$e(H, p) \leq F_r(x).$$

Proof. Let $q = 1 - p_b$. The restriction of c to $H - b$ is a fractional K_{r-1} -cover, since every K_{r-1} in $H - b$, together with b , forms a K_r .

If $q = 0$, then $x = e(H, p) = 0$. Suppose $q > 0$. If $q < (r-1)x$, then

$$e(H, p) \leq q(1-q) + \frac{q^2}{2} \leq B_{r,r-1}(x),$$

because $q - q^2/2$ is increasing on $[0, 1]$. If $q \geq (r-1)x$, normalize the remaining masses by q . The restricted cover has normalized mass $x/q \leq 1/(r-1)$, and the induction hypothesis and Lemma 2.3 give

$$e(H, p) \leq q(1-q) + q^2 F_{r-1}(x/q) \leq F_r(x).$$

□

3 Structure of a support-minimal weighted counterexample

3.1 The extremal setup

Fix $r \geq 3$ and assume inductively that Theorem 2.1 holds for clique size $r-1$. Suppose, for a contradiction, that it fails for clique size r . Since $F_r(1/r) = 1/2$ and every weighted graph has edge density at most $1/2$, there is a number $\bar{x} < 1/r$ for which a counterexample exists.

Among all counterexamples at $\bar{x}$, choose the smallest possible number m of positive-mass vertices. On a fixed labeled m -vertex set, consider all triples (H, p, c_0) in which H is a graph, p lies in the closed probability simplex, and $c_0 \in [0, 1]^{V(H)}$ is a fractional K_r -cover with $\langle p, c_0 \rangle \leq \bar{x}$. Maximize $e(H, p)$ over these triples. There are only finitely many choices for H , and for each one the feasible set is compact, so a maximum exists. The original counterexample shows that this maximum satisfies

$$e(H, p) > F_r(\bar{x}),$$

and a maximizing mass vector cannot lie on the boundary of the simplex: deleting a zero-mass vertex would preserve both the edge density and the cover constraint, producing a counterexample with smaller positive support. Fix a maximizing triple (H, p, c_0) ; in particular, every coordinate of p is positive.

Now keep (H, p) fixed and choose an extreme optimal solution c of the fractional-cover polyhedron

$$\mathcal{C}_r(H) = \{z \in \mathbb{R}_{\geq 0}^{V(H)} : z(K) \geq 1 \text{ for every } K \in \mathcal{K}_r(H)\}$$

that minimizes $\langle p, z \rangle$. Such a solution exists. Since every $p_v > 0$, an optimal cover cannot have a coordinate larger than one: truncating such a coordinate to one preserves all cover constraints and strictly lowers the

objective. Hence the entire optimal face is contained in the compact cube $[0, 1]^{V(H)}$. It is therefore a nonempty compact polytope and has an extreme point; because it is a face of $\mathcal{C}_r(H)$, this point is also extreme in the full cover polyhedron. Set

$$x = \langle p, c \rangle = \text{fcov}_r(H, p) \leq \bar{x}.$$

By monotonicity of F_r ,

$$e(H, p) > F_r(\bar{x}) \geq F_r(x). \quad (3.1)$$

All masses p_v are positive by the minimal choice of the support. Moreover, the support has at least three vertices: if it had at most two, then

$$e(H, p) \leq \frac{1}{4} \leq F_r(0) \leq F_r(x),$$

contrary to (3.1).

Lemma 3.1 (Positive cover coordinates are tight). *Every vertex v with $c_v > 0$ belongs to a K_r satisfying $c(K) = 1$.*

Proof. If every K_r containing v had cover sum strictly greater than one, then, because H is finite, one could decrease c_v by a positive amount while preserving all cover constraints. Since $p_v > 0$, this would lower the cover mass, contradicting the choice of c . $\square$

By maximality, H is saturated with respect to c : adding any nonedge destroys the cover property of c . Indeed, otherwise adding that edge would preserve the cover mass and strictly increase the edge density, contradicting the choice of the triple.

Lemma 3.2 (Saturation certificate). *For every nonedge uv of H , there is a copy $Q \cong K_{r-2}$ in $N_H(u) \cap N_H(v)$ such that*

$$c_u + c_v + c(Q) < 1.$$

Proof. After adding uv , every new K_r has the form $Q \cup \{u, v\}$ with $Q \cong K_{r-2}$ contained in the common neighborhood. Since c is no longer a cover after the edge is added, at least one such clique has cover sum less than one. $\square$

3.2 Strict negativity of the adjacency form

Let $M = A(H)$ be the adjacency matrix of H .

Lemma 3.3. *For every nonzero vector $z \in \mathbb{R}^{V(H)}$ satisfying*

$$\langle z, \mathbf{1} \rangle = 0, \quad \langle z, c \rangle = 0,$$

one has

$$z^\top M z < 0.$$

Proof. For sufficiently small positive and negative t , the vector $p + tz$ is a positive probability vector and

$$\langle p + tz, c \rangle = x \leq \bar{x}.$$

Hence it remains feasible for the extremal problem. Because that problem ranges over covers as well as mass vectors, the same cover c makes every such nearby vector an admissible choice. Thus p is an interior maximizer on this affine section, and the linear term vanishes:

$$z^\top M p = 0.$$

Thus

$$e(H, p + tz) = e(H, p) + \frac{t^2}{2} z^\top M z.$$

A positive quadratic term contradicts maximality. If the quadratic term is zero, move along the line until a coordinate of the mass vector first becomes zero. The edge density and cover mass remain unchanged, producing a counterexample with smaller support, again a contradiction. $\square$

Corollary 3.4. *The matrix M has at most two nonnegative eigenvalues. In particular, if $|V(H)| \geq 3$, then*

$$\lambda_3(M) < 0.$$

Proof. The subspace orthogonal to $\mathbf{1}$ and c has codimension at most two, and the quadratic form of M is strictly negative on this subspace. The min-max principle gives the conclusion. $\square$

We also record the first-order stationarity of the mass vector. Since p is positive and maximizes the edge density under the constraints

$$\langle p, \mathbf{1} \rangle = 1, \quad \langle p, c \rangle \leq \bar{x},$$

there are constants $\lambda \in \mathbb{R}$ and $\mu \geq 0$ such that

$$Mp = \lambda \mathbf{1} + \mu c. \tag{3.2}$$

Standing setup. Until the weighted induction is completed in [Section 6](#), the notation (H, p, c) refers to the extremal counterexample selected in [Section 3.1](#). Thus p has positive support, c is an extreme minimum fractional cover, and H is saturated with respect to c .

3.3 Co-chain support and the far-corner lemma

Lemma 3.5. *The independence number of H is at most two.*

Proof. If u, v, w were pairwise nonadjacent, then the three-dimensional coordinate subspace supported on $\{u, v, w\}$ would contain a nonzero vector z orthogonal to both $\mathbf{1}$ and c . Since this subspace contains no edges,

$$z^\top M z = 0,$$

contradicting [Lemma 3.3](#). $\square$

Let $J = \overline{H}$. Thus J is triangle-free.

Lemma 3.6. *The graph J is a chain graph. Equivalently, H is a co-chain graph.*

Proof. First we show that J is bipartite. Otherwise a shortest odd cycle in J is an induced cycle C_m with odd $m \geq 5$. Hence H contains the induced subgraph $\overline{C_m}$. The adjacency matrix of $\overline{C_m}$ has principal eigenvalue $m - 3$ and two further eigenvalues, both equal to

$$-1 + 2 \cos(\pi/m) > 0.$$

Cauchy interlacing would give $\lambda_3(M) > 0$, contradicting [Corollary 3.4](#).

Next, J cannot contain an induced $2K_2$, since its complement is an induced C_4 in H . The adjacency matrix of C_4 has third eigenvalue

$$\lambda_3(C_4) = 0.$$

Interlacing would give $\lambda_3(M) \geq 0$, again contradicting [Corollary 3.4](#). A bipartite graph with no induced $2K_2$ is a chain graph: on each side, incomparable neighborhoods would immediately produce an induced $2K_2$. $\square$

3.3.1 The sign rule on an induced path

Lemma 3.7 (P_4 sign rule). *Suppose $a - b - c - d$ is an induced P_4 in H . Then*

$$(c_c - c_a)(c_d - c_b) > 0.$$

Proof. Put

$$\alpha = c_c - c_a, \quad \beta = c_d - c_b.$$

If $(\alpha, \beta) \neq (0, 0)$, take the vector supported on the path with coordinates

$$z = (\beta, -\alpha, -\beta, \alpha).$$

Then $z \perp \mathbf{1}, c$ and

$$z^\top M z = -2\alpha\beta.$$

By [Lemma 3.3](#), this is strictly negative, so $\alpha\beta > 0$.

If $\alpha = \beta = 0$, the vector $(1, 1, -1, -1)$ is orthogonal to $\mathbf{1}$ and c , while its quadratic value on the path is $2 > 0$, another contradiction. $\square$

3.3.2 The Ferrers staircase

We shall use the following canonical form of a co-chain graph.

Lemma 3.8 (Canonical co-chain representation). *There are a (possibly empty) universal clique U and, unless H is complete, nonempty cliques*

$$L_0, L_1, \dots, L_{m-1} \quad \text{and} \quad R_1, R_2, \dots, R_m$$

such that $L = \bigcup_i L_i$ and $R = \bigcup_j R_j$ are cliques and

$$L_i R_j \subseteq E(H) \iff i \geq j. \quad (3.3)$$

The graph is complete exactly when $m = 0$.

Proof. Let $J = \overline{H}$. Remove the isolated vertices of J and place them in U ; these are precisely the universal vertices of H . If no vertices remain, then H is complete. Otherwise fix a bipartition $L \dot{\cup} R$ of the remaining chain graph. The neighborhoods of vertices in L are linearly ordered by inclusion. Merge equal-neighborhood vertices into twin classes and order the resulting distinct neighborhoods in strictly decreasing order under inclusion. Taking successive differences on the R side gives nonempty classes $R_1, \dots, R_m$ and twin classes $L_0, \dots, L_{m-1}$ such that

$$N_J(L_i) = R_{i+1} \cup \dots \cup R_m.$$

Equivalently, an $L_i R_j$ pair is an edge of H exactly when $i \geq j$. Since J is bipartite, both L and R are cliques in H . $\square$

Lemma 3.9 (Far-corner lemma). *Under the standing setup, the Ferrers staircase has at most one step; that is, $m \leq 1$.*

Proof. Assume $m \geq 2$. For $1 \leq i \leq m$, define the set of cover differences

$$D_i = \{c_r - c_\ell : r \in R_i, \ell \in L_{i-1}\}.$$

For $1 \leq i < m$, arbitrary choices

$$\ell_{i-1} \in L_{i-1}, \quad \ell_i \in L_i, \quad r_i \in R_i, \quad r_{i+1} \in R_{i+1}$$

induce the path

$$\ell_{i-1} - \ell_i - r_i - r_{i+1}.$$

By Lemma 3.7, every element of D_i has positive product with every element of D_{i+1} . Hence none of the D_i contains zero, and all D_i have the same strict sign.

Choose $u \in L_0$ and $v \in R_m$. They are nonadjacent, and their common neighborhood is exactly U . By Lemma 3.2, there is a copy $Q \cong K_{r-2}$ in U such that

$$c_u + c_v + c(Q) < 1. \quad (3.4)$$

Suppose first that all D_i are positive. Choose $w \in L_{m-1}$. Since $Q \cup \{u, w\}$ is a K_r ,

$$c_u + c_w + c(Q) \geq 1.$$

Together with (3.4), this gives $c_w > c_v$, while positivity of D_m gives $c_v > c_w$, a contradiction.

If all D_i are negative, choose $z \in R_1$. The clique $Q \cup \{z, v\}$ gives $c_z > c_u$, whereas negativity of D_1 gives $c_z < c_u$. This is again a contradiction. $\square$

Corollary 3.10. *The support graph H is either complete or admits a partition*

$$V(H) = A \dot{\cup} B \dot{\cup} C$$

into nonempty cliques such that B is complete to $A \cup C$ and A is anticomplete to C .

Proof. The case $m = 0$ is complete. When $m = 1$, take $A = L_0$, $C = R_1$, and $B = U$. To see that B is nonempty, choose $u \in L_0$ and $v \in R_1$. The saturation certificate for the nonedge uv produces a copy of K_{r-2} in their common neighborhood U , so in fact $|B| \geq r - 2$. $\square$

4 Complete supports and terminal zero sets

4.1 Complete supports

Suppose first that H is complete. Order the masses as

$$p_1 \geq p_2 \geq \cdots$$

and set $p_j = 0$ beyond the support.

Lemma 4.1 (Hypersimplex formula). *For a complete support,*

$$\text{til}_r(H, p) = \min_{0 \leq s \leq r-1} \frac{1 - \sum_{j=1}^s p_j}{r - s}.$$

Proof. Let a fractional tiling have size q , and let y_j be its load at vertex j . Every K_r uses each support vertex at most once, so

$$0 \leq y_j \leq \min\{p_j, q\}, \quad \sum_j y_j = rq.$$

Thus

$$\sum_j \min\{p_j, q\} \geq rq. \tag{4.1}$$

Conversely, the case $q = 0$ is trivial. If $q > 0$ and (4.1) holds, choose numbers $0 \leq y_j \leq \min\{p_j, q\}$ with total rq . The vector $(y_j/q)_j$ belongs to the hypersimplex

$$\left\{ z : 0 \leq z_j \leq 1, \sum_j z_j = r \right\},$$

which is the convex hull of the incidence vectors of r -subsets. Decomposing it into these incidence vectors gives a fractional K_r -tiling of size q .

For the ordered sequence (p_j) , condition (4.1) is equivalent to

$$q \leq \frac{1 - \sum_{j=1}^s p_j}{r - s} \quad (0 \leq s \leq r - 1).$$

Taking the largest feasible q proves the formula. □

Proposition 4.2. *If H is complete, then*

$$e(H, p) \leq F_r(x), \quad x = \text{fcov}_r(H, p).$$

Proof. If $x = 1/r$, then $e(H, p) \leq 1/2 = F_r(1/r)$, so assume $x < 1/r$. By (2.1) and Lemma 4.1, there is an $s \in \{0, \dots, r-1\}$ such that

$$1 - \sum_{j=1}^s p_j \leq (r - s)x.$$

The choice $s = 0$ would give $1 \leq rx$, contrary to $x < 1/r$. Hence $s \geq 1$. Put $i = r - s$. Then

$$\sum_{j=1}^{r-i} p_j \geq 1 - ix.$$

By Cauchy–Schwarz,

$$\sum_j p_j^2 \geq \frac{(1 - ix)^2}{r - i}.$$

Since H is complete, summing the weights over all unordered pairs gives

$$e(H, p) = \frac{1}{2} \left(1 - \sum_j p_j^2 \right) \leq B_{r,i}(x) \leq F_r(x).$$

□

The selected counterexample therefore has the one-step support of [Corollary 3.10](#). We write

$$V(H) = A \dot{\cup} B \dot{\cup} C, \quad (4.2)$$

where A, B, C are cliques, B is complete to both ends, and A is anticomplete to C . Each of A, B, C is a module: its vertices have the same neighbors outside the part. By saturation, B contains a copy of K_{r-2} ; in particular,

$$|B| \geq r - 2. \quad (4.3)$$

4.2 A weighted extension of Turán's theorem

We use the following weighted consequence of the theorem of Allen–Böttcher–Hladký–Piguet [\[ABHP14\]](#).

Theorem 4.3 (Weighted extension of Turán). *Let $r \geq 3$, let (G, p) be a weighted graph, and let $Z \subseteq V(G)$ have total mass z . If no copy of K_r intersects Z , then*

$$e(G, p) \leq \begin{cases} \frac{r-2}{2(r-1)}, & z \geq \frac{1}{r-1}, \\ \frac{1}{2} - z + \frac{r-1}{2} z^2, & 0 \leq z \leq \frac{1}{r-1}. \end{cases}$$

Proof. For rational masses, fix a common denominator D . For every positive integer q , replace each support vertex v by an independent class of size qDp_v . The hypothesis is preserved, and [\[ABHP14, Theorem 2\(a\)\]](#) gives the displayed asymptotic inequality after division by $(qD)^2$ and passage to the limit $q \rightarrow \infty$. Approximation by rational probability vectors proves the general case. $\square$

Let

$$Z = \{v : c_v = 0\}.$$

We call Z *terminal* if no copy of K_r intersects Z .

Lemma 4.4 (Zero-set mass). *One has*

$$p(Z) \geq 1 - rx.$$

Proof. Choose an optimal weighted fractional tiling t dual to c , and write

$$L(v) = \sum_{K \ni v} t_K$$

for its load at v . Complementary slackness gives

$$c_v > 0 \implies L(v) = p_v.$$

Hence

$$p(V(H) \setminus Z) = \sum_{c_v > 0} L(v) \leq \sum_v L(v) = r \sum_K t_K = rx.$$

$\square$

Lemma 4.5 (Terminal case). *If Z is terminal, then*

$$e(H, p) \leq F_r(x).$$

Proof. Put $z = p(Z)$ and $\ell = 1 - rx$. By [Lemma 4.4](#), $z \geq \ell$. If $z \geq 1/(r-1)$, then [Theorem 4.3](#) gives the Turán density

$$e(H, p) \leq \frac{r-2}{2(r-1)} \leq B_{r,1}(x).$$

If $z < 1/(r-1)$, then necessarily $\ell < 1/(r-1)$, so R_r is defined. The function

$$g(z) = \frac{1}{2} - z + \frac{r-1}{2} z^2$$

is decreasing on $[0, 1/(r-1)]$. Therefore

$$e(H, p) \leq g(z) \leq g(\ell) = R_r(x).$$

$\square$

4.3 A universal zero atom

Every vertex of B is universal in H . Thus [Lemma 2.4](#) gives the following immediate reduction.

Corollary 4.6. *In the selected support-minimal counterexample,*

$$c_b > 0 \quad (b \in B).$$

Proof. A zero-cover vertex in B would be a universal zero atom, and [Lemma 2.4](#) would contradict [\(3.1\)](#). $\square$

The terminal case has now been settled, and every vertex of B has positive cover weight. For completeness, the classification in the next section still records the terminal alternative.

5 Extreme covers on a one-step support

The crucial simplification is that an extreme minimum cover on the support [\(4.2\)](#) has at most two positive values.

For $S \subseteq V(H)$ with $|S| \geq r$, define

$$m_S(c) = \min\{c(K) : K \subseteq S, |K| = r\}.$$

We call S *active* if $m_S(c) = 1$. The two complete sides are

$$S_A = A \cup B, \quad S_C = B \cup C.$$

Lemma 5.1 (Tight sets use the lowest levels). *Let S be one of the two complete sides and let $K \subseteq S$ be a tight r -set, so $c(K) = 1$. If $u \in S \setminus K$ and $v \in K$, then*

$$c_u \geq c_v.$$

Consequently, every tight r -set in S has the same number of vertices at each cover level.

Proof. If $c_u < c_v$, then replacing v by u gives an r -set in the same complete side with cover sum less than one, contradicting feasibility. Thus every tight set consists of r vertices of minimum possible cover weight, with arbitrary choices only inside the threshold level. The number selected from each level is therefore fixed. $\square$

Lemma 5.2 (Two-level lemma). *Under the standing setup, the extreme minimum cover c has at most two distinct positive coordinate values.*

Moreover, if it has exactly two positive values, then the two active sides contribute linearly independent level-count equations.

Proof. Let

$$0 < \theta_1 < \dots < \theta_t$$

be the distinct positive values of c . For each active side S , let

$$a^S = (a_1^S, \dots, a_t^S)$$

be the vector counting how many vertices of each level occur in a tight r -set. This is well-defined by [Lemma 5.1](#).

If $t > 2$, choose a nonzero vector $z \in \mathbb{R}^t$ orthogonal to the at most two vectors a^{S_A}, a^{S_C} arising from active sides. Perturb every coordinate at level θ_j to $\theta_j \pm \varepsilon z_j$. Choose $\varepsilon > 0$ small enough that all positive coordinates remain positive, the strict order of the levels is unchanged, and every non-tight r -set remains feasible. Such a choice exists because there are only finitely many r -sets; if non-tight sets occur, take ε below their minimum positive slack divided by the maximum nonzero change of a constraint, while constraints whose change is zero impose no restriction. The sum of every tight r -set on an active side is unchanged, and every r -set on an inactive side remains strictly above one. Both perturbed vectors are therefore fractional covers, with c as their midpoint, contradicting the extremality of c .

The same perturbation shows that, when $t = 2$, the active level-count vectors must span $\mathbb{R}^2$. $\square$

We now combine this lemma with saturation to classify the nonterminal possibilities.
Let

$$m_B = |B|,$$

and order the B -weights as

$$b_1 \leq b_2 \leq \dots \leq b_{m_B}.$$

By (4.3), $m_B \geq r - 2$. Put

$$\sigma = \sum_{i=1}^{r-2} b_i. \quad (5.1)$$

Since every pair $a \in A$, $z \in C$ is a nonedge whose common neighborhood is B , Lemma 3.2 gives an $(r - 2)$ -clique $Q \subseteq B$ with $c_a + c_z + c(Q) < 1$. By the definition of σ , one has $\sigma \leq c(Q)$, and therefore

$$c_a + c_z + \sigma < 1 \quad (a \in A, z \in C). \quad (5.2)$$

Lemma 5.3. *Each of A and C contains at most one zero-cover vertex.*

Proof. If, say, $a_1, a_2 \in A$ were distinct zero-cover vertices, then $\{a_1, a_2\}$ together with the $r - 2$ lightest vertices of B would form a K_r . Hence $\sigma \geq 1$, contradicting (5.2). $\square$

Theorem 5.4 (Rigid cover classification). *Assume the standing setup and the partition (4.2), and suppose that $c_b > 0$ for every $b \in B$. After possibly interchanging A and C , one of the following holds.*

- (i) *No copy of K_r intersects the zero set Z .*
- (ii) *The sets A and C each consist of one zero-cover vertex, and*

$$c_b = \frac{1}{r-1} \quad (b \in B).$$

- (iii) *The set C consists of one zero-cover vertex. There are integers*

$$a_0 = |A|, \quad 0 \leq s \leq r - 2, \quad t \geq 0, \quad j = r - 1 - s \geq 1,$$

such that

$$c|_A \equiv \frac{1}{r},$$

exactly s vertices of B have cover value $1/r$, and the remaining t vertices have cover value

$$\gamma = \frac{j+1}{rj}.$$

Furthermore,

$$a_0 \geq j + 1, \quad t \geq j.$$

Proof. We divide the proof into several steps.

Step 1: the case $m_B = r - 2$. Suppose first that $m_B = r - 2$. If $|A|, |C| \geq 2$, let a_1, a_2 be the two lightest vertices of A and z_1, z_2 the two lightest vertices of C . The cliques formed by either pair together with all of B give

$$c_{a_1} + c_{a_2} + \sigma \geq 1, \quad c_{z_1} + c_{z_2} + \sigma \geq 1.$$

Hence

$$c_{a_2}, c_{z_2} \geq \frac{1 - \sigma}{2},$$

which contradicts (5.2) for the pair a_2, z_2 . Thus one end is a singleton; assume $C = \{z\}$.

The side $B \cup C$ has only $r - 1$ vertices, so z belongs to no K_r and optimality of c gives $c_z = 0$. If A contained a zero-cover vertex a and another vertex a' , then the clique $\{a, a'\} \cup B$ would give

$$c_{a'} + \sigma \geq 1,$$

contradicting (5.2) for a', z . Therefore no K_r intersects the zero set, and we are in case (i).

Step 2: preliminary consequences when $m_B \geq r - 1$. Assume now $m_B \geq r - 1$, and set

$$\beta = b_{r-1}.$$

For every $a \in A$ and $z \in C$, the $r - 1$ lightest vertices of B , together with a or z , form a K_r . Hence

$$c_a + \sigma + \beta \geq 1, \quad c_z + \sigma + \beta \geq 1.$$

Combining these inequalities with (5.2), we obtain

$$c_a < \beta, \quad c_z < \beta \quad (a \in A, z \in C). \quad (5.3)$$

Suppose some end vertex has positive cover weight. By Lemma 5.2, the positive levels are then

$$0 < \alpha < \gamma,$$

all positive end vertices have value α , and $\beta = \gamma$. Let

$$s = |\{b \in B : c_b = \alpha\}|.$$

Since the $(r - 1)$ st lightest B -weight is γ ,

$$s \leq r - 2. \quad (5.4)$$

For an end $E \in \{A, C\}$, write $z_E \in \{0, 1\}$ for the number of zero-cover vertices in E , and n_E for the number of positive vertices there. If the side $E \cup B$ is active, its tight level equation has one of the following two forms:

$$(r - z_E)\alpha = 1, \quad \text{if } n_E + s \geq r - z_E, \quad (5.5)$$

$$(n_E + s)\alpha + (r - z_E - n_E - s)\gamma = 1, \quad \text{if } n_E + s < r - z_E. \quad (5.6)$$

We call these the α -threshold and γ -threshold cases.

Step 3: both ends cannot contain positive vertices. Assume $n_A, n_C \geq 1$. Both sides are active by Lemma 3.1.

If both $A \cup B$ and $B \cup C$ are in the α -threshold case, then no tight clique contains a γ -vertex of B , contradicting Lemma 3.1.

Suppose $A \cup B$ is in the α -threshold case and $B \cup C$ is in the γ -threshold case. Comparing (5.5) and (5.6) gives

$$(z_C - z_A)\alpha = (r - z_C - n_C - s)(\gamma - \alpha) > 0.$$

Thus $z_A = 0$, $z_C = 1$, and $\alpha = 1/r$. Put

$$q = n_C \geq 1, \quad j = r - 1 - q - s \geq 1.$$

The tight equation for $B \cup C$ gives

$$\gamma = \frac{j+1}{rj}.$$

For positive vertices $a \in A$ and $z \in C$, the left-hand side of (5.2) is at least

$$\begin{aligned} 2\alpha + \sigma &= \frac{s+2}{r} + (r-2-s)\frac{j+1}{rj} \\ &= 1 + \frac{j+q-1}{rj} > 1, \end{aligned}$$

a contradiction. The reverse assignment of thresholds is symmetric.

It remains to suppose that both complete sides are in the γ -threshold case. If $z_A = z_C$, subtracting the two equations gives $n_A = n_C$, so the two level-count equations are identical. This contradicts the linear independence required by Lemma 5.2. If, say, $z_A = 0$ and $z_C = 1$, subtraction gives

$$(n_A - n_C)(\gamma - \alpha) = \gamma.$$

In particular $n_A \geq 2$, and comparison of the tight equation for $A \cup B$ with $2\alpha + \sigma$ yields

$$2\alpha + \sigma - 1 = (n_A - 2)(\gamma - \alpha) \geq 0,$$

again contradicting (5.2). Thus both ends cannot contain positive vertices.

Step 4: one end is entirely zero. After interchanging the ends, assume every vertex of C has cover value zero. By Lemma 5.3,

$$C = \{z\}.$$

Suppose first that A contains both a zero vertex and a positive vertex. Then A has exactly one zero vertex and all its positive vertices have value α . The side $A \cup B$ is active. If it is in the α -threshold case, then

$$(r - 1)\alpha = 1.$$

A γ -vertex of B is not contained in a tight r -set in $A \cup B$, so $B \cup C$ must be active. Its tight equation is

$$s\alpha + (r - 1 - s)\gamma = 1,$$

which forces $\gamma = \alpha$, a contradiction. If $A \cup B$ is in the γ -threshold case and $B \cup C$ is active, subtracting their equations gives

$$n_A(\alpha - \gamma) = 0,$$

again impossible. If $B \cup C$ is inactive, there is only one active level-count equation for the two positive levels, contradicting Lemma 5.2. Hence A cannot contain both zero and positive vertices.

If every vertex of A is zero, then Lemma 5.3 gives $A = \{a\}$. The two sides have the same multiset of cover values: one zero followed by the values on B . Consequently they have the same minimum r -set sum. By Lemma 3.1, at least one side is active, and hence both are active. If two positive levels occurred, their active level-count equations would be identical, contradicting Lemma 5.2. Thus all vertices of B have one common value λ . A tight clique exists by Lemma 3.1, and necessarily consists of one zero end and $r - 1$ vertices of B . Therefore

$$(r - 1)\lambda = 1,$$

and we obtain case (ii).

Finally, suppose every vertex of A is positive. Write $a_0 = |A|$; every A -vertex has value α . If $a_0 + s \geq r$, the side $A \cup B$ is in the α -threshold case, so

$$r\alpha = 1.$$

The γ -vertices of B require $B \cup C$ to be active, and

$$s\alpha + (r - 1 - s)\gamma = 1.$$

Putting $j = r - 1 - s$, we obtain

$$\alpha = \frac{1}{r}, \quad \gamma = \frac{j + 1}{rj}.$$

The inequality $a_0 + s \geq r$ is exactly $a_0 \geq j + 1$, and $m_B = s + t \geq r - 1$ gives $t \geq j$. This is case (iii).

It remains to rule out $a_0 + s < r$. Then $A \cup B$ is in the γ -threshold case. If $B \cup C$ were inactive, the two-level cover would have only one active count equation, contrary to Lemma 5.2. Thus $B \cup C$ is active, and subtracting the two γ -threshold equations gives

$$a_0\alpha + (1 - a_0)\gamma = 0.$$

For $a_0 = 1$ this gives $\alpha = 0$. For $a_0 \geq 2$, perturb two A -coordinates by $+\varepsilon$ and $-\varepsilon$. Since $a_0 + s < r$, every tight r -set in $A \cup B$ contains all vertices of A , while tight r -sets in $B \cup C$ contain none of them, so all tight sums remain unchanged. Because there are finitely many non-tight r -sets, they have a positive minimum slack; choosing ε below that slack ensures that both perturbations remain covers. This contradicts extremality of c and completes the classification. $\square$

5.1 The two rigid nonterminal types

We evaluate cases (ii) and (iii) of Theorem 5.4.

It is convenient to use the deficiency

$$\Delta(H, p) := 1 - 2e(H, p). \tag{5.7}$$

For a simple weighted support, this equals the sum of the diagonal atom terms $\sum_v p_v^2$ and twice the total mass of the nonedges between distinct atoms.

5.1.1 Two zero ends

Proposition 5.5. *Suppose A and C each consist of one zero-cover vertex and $c_b = 1/(r-1)$ for all $b \in B$. Then*

$$e(H, p) \leq B_{r, r-1}(x).$$

Proof. Let $b = p(B)$. Then

$$x = \frac{b}{r-1}, \quad p(A) + p(C) = 1 - (r-1)x.$$

Since the only cross-module nonedges are between A and C ,

$$\begin{aligned} \Delta(H, p) &= p(A)^2 + p(C)^2 + 2p(A)p(C) + \sum_{v \in B} p_v^2 \\ &\geq (1 - (r-1)x)^2. \end{aligned}$$

This is equivalent to the claimed $B_{r, r-1}$ bound. $\square$

5.1.2 The rigid two-level type

Assume case (iii) of [Theorem 5.4](#). Thus

$$C = \{z\}, \quad c_z = 0,$$

all $a_0 = |A|$ vertices of A have cover value $1/r$, and B consists of

- s low vertices of cover value $1/r$;
- t high vertices of cover value

$$\gamma = \frac{j+1}{rj}, \quad j = r-1-s.$$

We have

$$a_0 \geq j+1, \quad t \geq j. \tag{5.8}$$

Put

$$N = a_0 - 1, \quad h = N - j = a_0 - 1 - j \geq 0, \quad \kappa = t - j \geq 0. \tag{5.9}$$

By the stationarity equation (3.2), atoms in the same module and at the same cover level have equal masses. Indeed, subtracting their KKT equations shows that the difference of their weighted degrees, which is the negative of the difference of their own masses, is zero. Write

w = the mass of each vertex of A ,

L = the mass of each low vertex of B ($s > 0$),

u = the mass of each high vertex of B , $d = p_z$.

All masses corresponding to actual atoms are positive; the symbol L is needed only when $s > 0$.

Lemma 5.6 (KKT parametrization). *Define*

$$v = \frac{N}{j}w.$$

When $s = 0$, set $L = u + v$ formally; every expression involving L below is multiplied by s . Then

$$w = \frac{j}{N}v, \quad d = u + \frac{h}{N}v, \quad L = u + v, \tag{5.10}$$

and

$$rv + (r + \kappa)u = 1. \tag{5.11}$$

Moreover, if $\ell = 1 - rx$, then

$$\ell = \frac{h}{N}v - \frac{\kappa}{j}u. \tag{5.12}$$

Finally,

$$\Delta(H, p) = (r + \kappa)u^2 + 2ruv + \left(r + \frac{jh}{N}\right)v^2. \tag{5.13}$$

Proof. The weighted degrees of an A -vertex, a B -vertex of mass q , and the vertex z are respectively

$$1 - d - w, \quad 1 - q, \quad p(B).$$

We shall use the total-mass identity

$$a_0 w + sL + tu + d = 1. \quad (5.14)$$

Equation (3.2) and the comparison of an A -vertex with z give

$$\frac{\mu}{r} = Nw. \quad (5.15)$$

Comparing a high B -vertex with z , and using the total-mass identity, gives

$$a_0 w + d - u = \mu\gamma = Nw \frac{j+1}{j},$$

so

$$d - u = \frac{h}{j}w.$$

If $s > 0$, comparison of a low B -vertex with z gives $L = d + w$. Substituting $v = Nw/j$ yields (5.10).

Substituting (5.10), $s = r - 1 - j$, and $t = j + \kappa$ into (5.14) gives (5.11).

The cover mass is

$$x = \frac{a_0 w + sL}{r} + t \frac{j+1}{rj} u.$$

Subtracting rx from (5.11) gives (5.12).

Finally, the missing edges consist of the atom diagonals and all pairs between A and z . Hence

$$\Delta(H, p) = a_0 w^2 + sL^2 + tu^2 + d^2 + 2a_0 wd.$$

Substitution of (5.10) gives (5.13). □

Proposition 5.7. *In the rigid two-level type,*

$$e(H, p) \leq R_r(x).$$

Proof. By definition, $\ell = 1 - rx$, so $x \leq 1/r$ gives $\ell \geq 0$. The inequalities $t \geq j \geq 1$ show that high vertices occur, so their common mass u is positive. Therefore (5.11) gives $v < 1/r$, and

$$\ell \leq \frac{h}{N}v < \frac{1}{r} < \frac{1}{r-1}.$$

Thus $R_r(x)$ is defined. It is enough to prove

$$\Delta(H, p) \geq 2\ell - (r-1)\ell^2. \quad (5.16)$$

Using $1 = rv + (r + \kappa)u$ from (5.11), the difference between the two sides of (5.16) is the homogeneous quadratic form

$$\mathcal{A}u^2 + 2\mathcal{B}uv + \mathcal{C}v^2, \quad (5.17)$$

where

$$\mathcal{A} = \frac{j^2(r + \kappa) + 2j\kappa(r + \kappa) + \kappa^2(r - 1)}{j^2}, \quad (5.18)$$

$$\mathcal{B} = \frac{jr(j + \kappa) - h\kappa(j - 1)}{j(j + h)}, \quad (5.19)$$

$$\mathcal{C} = \frac{j^2r + hj^2 + h^2(j - 1)}{(j + h)^2}. \quad (5.20)$$

Clearly $\mathcal{A}, \mathcal{C} > 0$. A direct factorization gives

$$\begin{aligned} \mathcal{A}\mathcal{C} - \mathcal{B}^2 &= \frac{hj\kappa + hjr + h\kappa r + j\kappa r}{j^2(j + h)^2} \\ &\quad \cdot \left(h(j - 1)(j + \kappa) + j(j^2 + \kappa(2j - 1)) \right). \end{aligned} \quad (5.21)$$

Both factors are nonnegative, and the second is strictly positive. Hence the matrix of (5.17) is positive semidefinite, proving (5.16). Since

$$1 - 2R_r(x) = 2\ell - (r - 1)\ell^2,$$

the result follows. $\square$

6 Completion of the weighted induction

We now prove Theorem 2.1 for clique size r , completing the induction.

Proof of Theorem 2.1. The case $r = 2$ is Proposition 2.2. Let $r \geq 3$ and assume the theorem for clique size $r - 1$. Suppose a counterexample exists and choose the extremal triple from Section 3.

By Corollary 3.10, the support is complete or a clique blow-up of P_3 . The complete case is ruled out by Proposition 4.2. Thus we have the partition (4.2).

If no K_r intersects the zero set, Lemma 4.5 gives the desired bound. If a vertex of B has zero cover value, Lemma 2.4 gives the desired bound. We may therefore apply Theorem 5.4. Case (i) is terminal, case (ii) is handled by Proposition 5.5, and case (iii) is handled by Proposition 5.7. Each alternative contradicts (3.1).

Thus no counterexample exists, and the induction is complete for every $r \geq 2$. If c is any fractional K_r -cover, then $\text{fcov}_r(H, p) \leq \langle p, c \rangle$; the final assertion of Theorem 2.1 therefore follows from the monotonicity in Lemma 1.5. $\square$

7 Graphons and finite graphs

We finish the proof of Theorem 1.1. We use the graphon tiling theory of Hladký, Hu, and Piguet [HHP21], together with the standard compactness theory for dense graph limits [LS06]. Let $W : \Omega^2 \rightarrow [0, 1]$ be a symmetric measurable function on an atomless probability space (Ω, μ) , and write

$$e(W) = \frac{1}{2} \int_{\Omega^2} W(x, y) \, d\mu(x) \, d\mu(y).$$

We recall the normalization of [HHP21]. A graphon fractional K_r -tiling is a measurable function $t : \Omega^r \rightarrow [0, \infty)$ that vanishes almost everywhere on tuples satisfying $\prod_{i < j} W(x_i, x_j) = 0$ and, for almost every $x \in \Omega$, obeys

$$\sum_{i=1}^r \int_{\Omega^{r-1}} t(x_1, \dots, x_{i-1}, x, x_{i+1}, \dots, x_r) \prod_{j \neq i} d\mu(x_j) \leq 1.$$

Its size is $\int_{\Omega^r} t \, d\mu^r$, and $\text{til}(K_r, W)$ is the supremum of these sizes. A graphon fractional K_r -cover is a measurable function $c : \Omega \rightarrow [0, 1]$ such that

$$\sum_{i=1}^r c(x_i) \geq 1$$

for μ^r -almost every $(x_1, \dots, x_r)$ with $\prod_{i < j} W(x_i, x_j) > 0$; its mass is $\int_{\Omega} c \, d\mu$, and the infimum of these masses is denoted by $\text{fcov}(K_r, W)$. Graphon tiling-cover duality and attainment of the cover minimum [HHP21, Theorem 3.16 and (3.7)] give

$$\text{til}(K_r, W) = \text{fcov}(K_r, W), \tag{7.1}$$

and the lower-semicontinuity theorem [HHP21, Theorem 3.4] gives

$$\text{til}(K_r, W) \leq \liminf_{n \rightarrow \infty} \frac{\nu(K_r, G_n)}{|V(G_n)|} \tag{7.2}$$

whenever G_n converges to W in cut distance.

Theorem 7.1 (Graphon density theorem). *For every $r \geq 3$ and every graphon W , if*

$$x = \text{til}(K_r, W),$$

then

$$e(W) \leq F_r(x).$$

Proof. By (7.1), choose a minimum fractional cover $c : \Omega \rightarrow [0, 1]$ with $\int_{\Omega} c \, d\mu = x$. If $x = 1/r$, then

$$e(W) \leq \frac{1}{2} = F_r(1/r),$$

so assume $x < 1/r$.

Take independent samples $X_1, X_2, \dots$ from μ . Conditional on the points, independently join every pair $i < j$ with probability $W(X_i, X_j)$, thereby defining an infinite random graph on $\mathbb{N}$; let H_m be its subgraph induced by $[m]$. With probability one,

$$\frac{e(H_m)}{m^2} \rightarrow e(W), \quad x_m := \frac{1}{m} \sum_{i=1}^m c(X_i) \rightarrow x.$$

Moreover, with probability one, for every m the sampled vector

$$c^{(m)} = (c(X_1), \dots, c(X_m))$$

is a fractional K_r -cover of H_m . Indeed, fix distinct indices $i_1, \dots, i_r$ and let

$$E = \left\{ (x_1, \dots, x_r) : \sum_{a=1}^r c(x_a) < 1 \right\}.$$

The graphon cover property says that

$$\mathbf{1}_E(x_1, \dots, x_r) \prod_{1 \leq a < b \leq r} W(x_a, x_b) = 0 \quad \text{for almost every } (x_1, \dots, x_r).$$

Consequently, the probability that this fixed index set is simultaneously a realized clique and violates the sampled cover inequality is zero. Taking the countable union over all choices of distinct indices, and hence over all m , proves the assertion.

For all sufficiently large m , $x_m < 1/r$. Apply Theorem 2.1 to the uniformly weighted graph H_m and the cover $c^{(m)}$:

$$\frac{e(H_m)}{m^2} \leq F_r(x_m).$$

Letting $m \rightarrow \infty$ and using Lemma 1.5 proves the theorem. $\square$

Proof of Theorem 1.1. The lower bound is Proposition 1.4. Suppose the upper bound fails uniformly. Then there are $\varepsilon > 0$, integers $n_j \rightarrow \infty$, integers $0 \leq k_j \leq n_j/r$, and graphs G_j on n_j vertices such that

$$\nu(K_r, G_j) \leq k_j$$

and

$$e(G_j) \geq (F_r(k_j/n_j) + \varepsilon)n_j^2.$$

Passing to a subsequence, assume $k_j/n_j \rightarrow x$ and, by compactness of graphon space [LS06], that G_j converges in cut distance to a graphon W . By (7.2),

$$\text{til}(K_r, W) \leq \liminf_{j \rightarrow \infty} \frac{\nu(K_r, G_j)}{n_j} \leq x.$$

By Theorem 7.1 and monotonicity of F_r ,

$$e(W) \leq F_r(\text{til}(K_r, W)) \leq F_r(x).$$

On the other hand, edge density is continuous under cut convergence, and Lemma 1.5 gives

$$e(W) = \lim_{j \rightarrow \infty} \frac{e(G_j)}{n_j^2} \geq F_r(x) + \varepsilon,$$

a contradiction. $\square$

8 The cases $r = 3$ and $r = 4$

The unified maximum in (1.4) reproduces the known density functions for triangles and for K_4 [ABHP15, HHLZ25].

Corollary 8.1 (Triangles). *For $0 \leq x \leq 1/3$,*

$$F_3(x) = \begin{cases} \frac{1}{4} + \frac{x}{2} - \frac{x^2}{4}, & 0 \leq x \leq \frac{2}{9}, \\ \frac{1}{4} + 2x^2, & \frac{2}{9} \leq x \leq \frac{1}{4}, \\ 2x - 2x^2, & \frac{1}{4} \leq x \leq \frac{5 + \sqrt{3}}{22}, \\ \frac{1}{2} - 3x + 9x^2, & \frac{5 + \sqrt{3}}{22} \leq x \leq \frac{1}{3}. \end{cases}$$

Consequently, uniformly for $0 \leq k \leq n/3$,

$$\text{ex}(n, (k+1)K_3) = F_3(k/n)n^2 + o(n^2).$$

Proof. The four curves in (1.4) are

$$\begin{aligned} B_{3,1}(x) &= \frac{1}{4} + \frac{x}{2} - \frac{x^2}{4}, & A_3(x) &= \frac{1}{4} + 2x^2, \\ B_{3,2}(x) &= 2x - 2x^2, & R_3(x) &= \frac{1}{2} - 3x + 9x^2. \end{aligned}$$

The successive relevant intersections are

$$\begin{aligned} B_{3,1} = A_3 &\iff x = \frac{2}{9}, & A_3 = B_{3,2} &\iff x = \frac{1}{4}, \\ B_{3,2} = R_3 &\iff x = \frac{5 + \sqrt{3}}{22}. \end{aligned}$$

Direct subtraction shows the indicated curve dominates throughout each intervening interval. The finite-graph assertion is [Theorem 1.1](#). $\square$

Corollary 8.2 (Cliques of size four). *For $0 \leq x \leq 1/4$,*

$$F_4(x) = \begin{cases} \frac{1}{3} + \frac{x}{3} - \frac{x^2}{6}, & 0 \leq x \leq \frac{2}{13}, \\ \frac{1}{3} + 2x^2, & \frac{2}{13} \leq x \leq \frac{1}{6}, \\ \frac{1}{4} + x - x^2, & \frac{1}{6} \leq x \leq \frac{4 - \sqrt{2}}{14}, \\ 3x - \frac{9}{2}x^2, & \frac{4 - \sqrt{2}}{14} \leq x \leq \frac{11 + \sqrt{7}}{57}, \\ 1 - 8x + 24x^2, & \frac{11 + \sqrt{7}}{57} \leq x \leq \frac{1}{4}. \end{cases}$$

Consequently, uniformly for $0 \leq k \leq n/4$,

$$\text{ex}(n, (k+1)K_4) = F_4(k/n)n^2 + o(n^2).$$

Proof. The five curves are

$$\begin{aligned} B_{4,1}(x) &= \frac{1}{3} + \frac{x}{3} - \frac{x^2}{6}, & A_4(x) &= \frac{1}{3} + 2x^2, \\ B_{4,2}(x) &= \frac{1}{4} + x - x^2, & B_{4,3}(x) &= 3x - \frac{9}{2}x^2, \\ R_4(x) &= 1 - 8x + 24x^2. \end{aligned}$$

The successive relevant intersections occur at

$$\frac{2}{13}, \quad \frac{1}{6}, \quad \frac{4 - \sqrt{2}}{14}, \quad \frac{11 + \sqrt{7}}{57},$$

respectively. Pairwise subtraction verifies dominance on the displayed intervals. The finite-graph assertion follows from [Theorem 1.1](#). $\square$

9 Concluding remarks

The argument gives a single density proof for every clique size $r \geq 3$. For $r = 3$ and $r = 4$ it recovers, with an $o(n^2)$ error, the previously known formulas of Allen–Böttcher–Hladký–Piguet and Hou–Hu–Liu–Zhang, respectively. For every $r \geq 5$ it proves the $o(n^2)$ -error form of Conjecture 8.1 from [\[HHLZ25\]](#). The induction starts from the Erdős–Gallai matching theorem at $r = 2$, rather than using the known triangle or K_4 density theorems as base cases.

The stronger $O(n)$ error known for $r = 3, 4$ and conjectured for general r is not addressed here. Obtaining it would require a finite stability and rounding theory for the weighted extremizers. Equality and near-equality in the adjacency-form inequality of [Lemma 3.3](#), the far-corner lemma, the two-level classification, and the positive-semidefinite form in [Proposition 5.7](#) suggest a possible route.

The extreme-cover step may also be useful beyond cliques. The argument suggests that an analogous bounded-level conclusion may hold when a support graph is the union of a bounded number of complete host regions. In the present problem, the co-chain reduction leaves exactly two complete sides, which is why the cover collapses to two levels.

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Declaration on the use of AI

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A Selected algebraic identities

For reference, we record two calculations used in the proof.

A.1 The determinant in the rigid type

Starting from

$$\ell = \frac{h}{j+h}v - \frac{\kappa}{j}u,$$

$$\Delta = (r + \kappa)u^2 + 2ruv + \left(r + \frac{jh}{j+h}\right)v^2,$$

and

$$1 = rv + (r + \kappa)u,$$

one obtains

$$\Delta - (2\ell - (r-1)\ell^2) = \mathcal{A}u^2 + 2\mathcal{B}uv + \mathcal{C}v^2,$$

with coefficients (5.18)–(5.20). Expanding and collecting terms gives

$$\mathcal{A}\mathcal{C} - \mathcal{B}^2 = \frac{hj\kappa + hjr + h\kappa r + j\kappa r}{j^2(j+h)^2} \cdot \left(h(j-1)(j+\kappa) + j(j^2 + \kappa(2j-1))\right).$$

The factorization is nonnegative because $h, \kappa \geq 0$ and $j \geq 1$ make both factors nonnegative.

A.2 The R -branch in the recursion

For $r = 4$, the transformed R_3 branch is

$$\Psi_x(q) = (1 - 3x)q - \frac{1}{2}q^2 + 9x^2, \quad 3x \leq q \leq \min\{6x, 1\}.$$

Its interior stationary value on $1/9 \leq x \leq 1/6$ is

$$\Psi_x(1 - 3x) = \frac{1}{2} - 3x + \frac{27}{2}x^2,$$

and

$$A_4(x) - \Psi_x(1 - 3x) = -\frac{1}{6} + 3x - \frac{23}{2}x^2.$$

The latter is concave and is positive at both endpoints $1/9$ and $1/6$.

For $r \geq 5$, let

$$y_0 = \frac{r-3}{(r-1)(r-2)}.$$

At the left endpoint of the R_{r-1} domain,

$$B_{r-1,r-2}(y_0) - R_{r-1}(y_0) = \frac{(r-3)^2}{2(r-2)(r-1)^2}.$$

At the endpoint $q = 1$ of the peeling interval,

$$B_{r,r-1}(x) - R_{r-1}(x) = \frac{(1 - (r-1)x)((r-1)^2x - r + 3)}{2}.$$

These are the endpoint comparisons used in [Lemma 2.3](#).